

Student Number _____



ABBOTSLEIGH

AUGUST 2007

YEAR 12

ASSESSMENT 4

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks – 120

- Attempt Questions 1-8.
- All questions are of equal value.
- Answer each question in a new booklet.

Outcomes assessed

HSC course

- E1** appreciates the creativity, power and usefulness of mathematics to solve a broad range of problems
- E2** chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings
- E3** uses the relationship between algebraic and geometric representations of complex numbers and of conic sections
- E4** uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials
- E5** uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces and resisted motion
- E6** combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions
- E7** uses the techniques of slicing and cylindrical shells to determine volumes
- E8** applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems
- E9** communicates abstract ideas and relationships using appropriate notation and logical argument

Harder applications of the Extension 1 Mathematics course are included in this course. Thus the Outcomes from the Extension 1 Mathematics course are included.

From the Extension 1 Mathematics Course

Preliminary course

- PE1** appreciates the role of mathematics in the solution of practical problems
- PE2** uses multi-step deductive reasoning in a variety of contexts
- PE3** solves problems involving inequalities, polynomials, circle geometry and parametric representations
- PE4** uses the parametric representation together with differentiation to identify geometric properties of parabolas
- PE5** determines derivatives that require the application of more than one rule of differentiation
- PE6** makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations

HSC course

- HE1** appreciates interrelationships between ideas drawn from different areas of mathematics
- HE2** uses inductive reasoning in the construction of proofs
- HE3** uses a variety of strategies to investigate mathematical models of situations involving projectiles, simple harmonic motion or exponential growth and decay
- HE4** uses the relationship between functions, inverse functions and their derivatives
- HE5** applies the chain rule to problems including those involving velocity and acceleration as functions of displacement
- HE6** determines integrals by reduction to a standard form through a given substitution
- HE7** evaluates mathematical solutions to problems and communicates them in an appropriate form

Total marks – 120

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

QUESTION 1 (15 marks)

Use a SEPARATE writing booklet.

(a) Find $\int \sin^3 \theta d\theta$. 2

(b) (i) Express $\frac{3x+1}{(x+1)(x^2+1)}$ in the form $\frac{a}{x+1} + \frac{bx+c}{x^2+1}$. 2

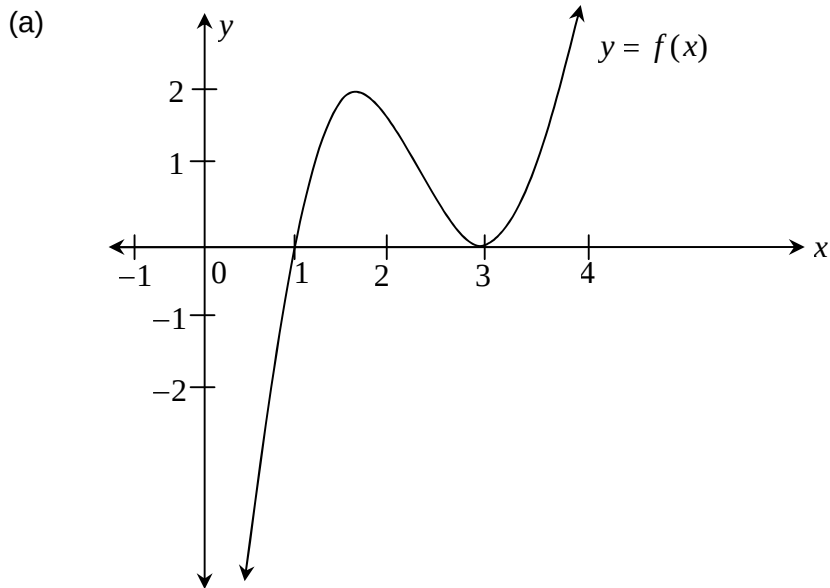
(ii) Hence find $\int \frac{3x+1}{(x+1)(x^2+1)}$. 2

(c) Use the substitution $x = 2 \sin \theta$, or otherwise, to evaluate $\int_1^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx$. 3

(d) Find $\int x^2 \sqrt{3-x} dx$. 3

(e) Evaluate $\int_0^1 \tan^{-1} \theta d\theta$. 3

QUESTION 2 (15 marks)
Start a new writing booklet.



The diagram above is a sketch of the function $y = f(x)$.

On separate diagrams sketch:

(i) $y = (f(x))^2$ 2

(ii) $y = \sqrt{f(x)}$ 2

(iii) $y = \ln[f(x)]$ 2

(iv) $y^2 = f(x)$ 2

(b) (i) If $f'(x) = \frac{2-x}{x^2}$ and $f(1) = 0$, find $f''(x)$ and $f(x)$. 3

(ii) Explain why the graph of $f(x)$ has only one turning point and find the value of the function at that point, stating whether it is a maximum or a minimum value. 2

(iii) Show that $f(4)$ and $f(5)$ have opposite signs and draw a sketch of $f(x)$. 2

QUESTION 3 (15 marks)
Start a new writing booklet.

(a) Express $(\sqrt{3} + i)^8$ in the form $x + iy$. 3

(b) On an Argand diagram, sketch the region where the inequalities 3

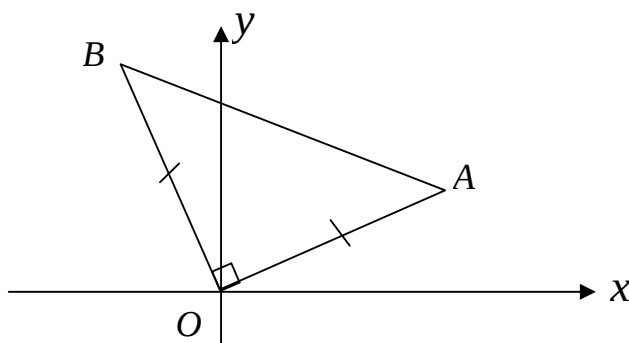
$$|z| \leq 3 \text{ and } -\frac{2\pi}{3} \leq \arg(z + 2) \leq \frac{\pi}{6} \text{ both hold.}$$

(c) Show that $\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} = \sin \theta + i \cos \theta$. 3

(d) (i) Express $z = \frac{-1 + i}{\sqrt{3} + i}$ in modulus-argument form. 2

(ii) Hence evaluate $\cos \frac{7\pi}{12}$ in surd form. 2

(e) The Argand diagram below shows the points A and B which represent the complex numbers z_1 and z_2 respectively.



Given that $\triangle BOA$ is a right-angled isosceles triangle, show that $(z_1 + z_2)^2 = 2z_1z_2$. 2

QUESTION 4 (15 marks)
Start a new writing booklet.

- (a) If $z = 1 + i$ is a root of the equation $z^3 + pz^2 + qz + 6 = 0$ where p and q are real, find p and q . **3**

- (b) Show that if the polynomial $f(x) = x^3 + px + q$ has a multiple root, then $4p^3 + 27q^2 = 0$. **3**

- (c) The base of a solid is the region in the first quadrant bounded by the curve $y = \sin x$, the x -axis and the line $x = \frac{\pi}{2}$. **3**

Find the volume of the solid if every cross-section perpendicular to the base and the x -axis is a square.

- (d) (i) Find the five roots of the equation $z^5 = 1$. Give the roots in modulus-argument form. **2**

- (ii) Show that $z^5 - 1$ can be factorised in the form :

$$z^5 - 1 = (z - 1)\left(z^2 - 2z \cos \frac{2\pi}{5} + 1\right)\left(z^2 - 2z \cos \frac{4\pi}{5} + 1\right) \quad \textbf{2}$$

- (iii) Hence show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$. **2**

QUESTION 5 (15 marks)

Start a new writing booklet.

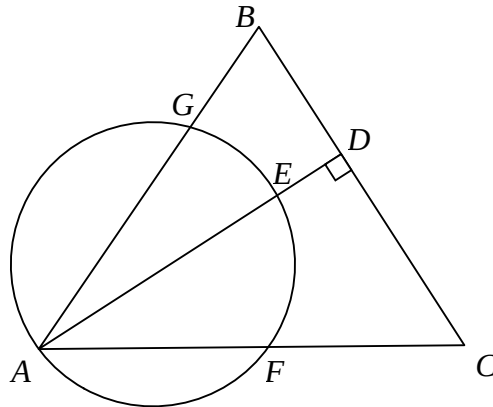
- (a) The ellipse $(x-1)^2 + \frac{y^2}{4} = 1$ is rotated about the y -axis.

Use the method of slicing to find the volume of the solid formed by the rotation.

4

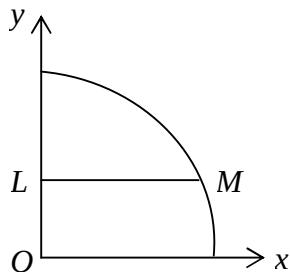
- (b) In the triangle ABC , AD is the perpendicular from A to BC . E is any point on AD and the circle drawn with AE as diameter cuts AC at F and AB at G .

4



Prove B, G, F and C are concyclic.

- (c) The diagram below shows the part of the circle $x^2 + y^2 = a^2$ in the first quadrant.



- (i) If the horizontal line LM through $L(0, b)$, where $0 < b < a$, divides the area between the curve and the coordinates axes into two equal parts, show that

$$\sin^{-1} \frac{b}{a} + \frac{b\sqrt{a^2 - b^2}}{a^2} = \frac{\pi}{4}.$$

3

- (ii) If the radius of the circle is 1 unit, show that b can be found by solving the equation

$$\sin 2\theta = \frac{\pi}{2} - 2\theta, \text{ where } \theta = \sin^{-1} b.$$

3

- (iii) Without attempting to solve the equation, how could θ (and hence b) be approximated?

1

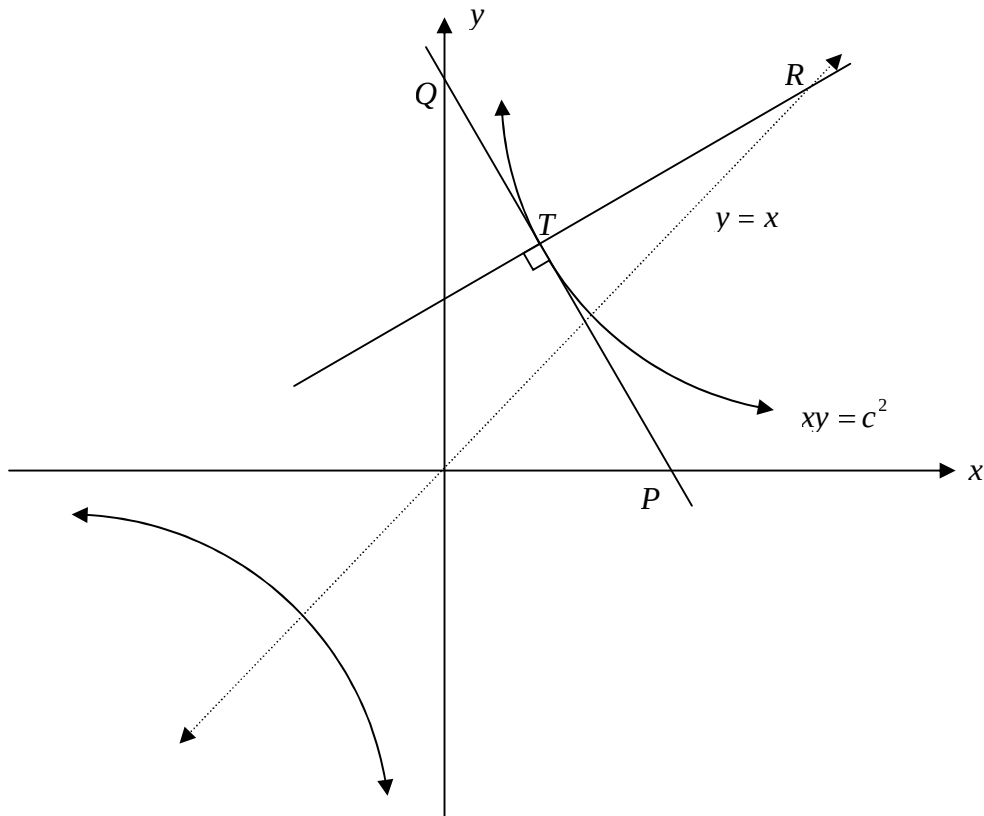
QUESTION 6 (15 marks)

Start a new writing booklet.

- (a) An ellipse has equation $\frac{x^2}{4} + \frac{y^2}{3} = 1$ with vertices $A(2,0)$ and $A'(-2,0)$. P is a point (x_1, y_1) on the ellipse.
- (i) Find its eccentricity, coordinates of its foci, S and S' , and the equations of its directrices. **3**
- (ii) Prove that the sum of the distances SP and $S'P$ is independent of the position of P . **2**
- (iii) Show that the equation of the tangent to the ellipse at P is $\frac{xx_1}{4} + \frac{yy_1}{3} = 1$. **2**
- (iv) The tangent at $P(x_1, y_1)$ meets the directrix at T . Prove that angle PST is a right angle. **3**
- (b) If $a + b + c = 1$,
- (i) Prove $a^2 + b^2 \geq 2ab$. **1**
- (ii) Prove $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$. **2**
- (iii) Prove $(1-a)(1-b)(1-c) \geq 8abc$. **2**

QUESTION 7 (15 marks)
Start a new writing booklet.

- (a) The point $T(ct, \frac{c}{t})$ lies on the hyperbola $xy = c^2$.
 The tangent at T meets the x -axis at P and the y -axis at Q .
 The normal at T meets the line $y = x$ at R .



*not to
scale*

You may assume that the tangent at T has equation $x + t^2y = 2ct$.

- (i) Find the coordinates of P and Q . 2
- (ii) Find the equation of the normal at T . 2
- (iii) Show that the x -coordinate of R is $x = \frac{c}{t}(t^2 + 1)$. 2
- (iv) Prove that $\triangle PQR$ is isosceles. 3
- (b) (i) If $I_n = \int \frac{dx}{(x^2 + 1)^n}$ prove that $I_n = \frac{1}{2(n-1)} \left[\frac{x}{(x^2 + 1)^{n-1}} + (2n-3)I_{n-1} \right]$. 4
- (ii) Hence evaluate $\int_0^1 \frac{dx}{(x^2 + 1)^2}$. 2

QUESTION 8 (15 marks)
Start a new writing booklet.

Marks

- (a) A plane of mass M kg on landing, experiences a variable resistive force due to air resistance of magnitude Bv^2 newtons, where v is the speed of the plane. That is, $M\ddot{x} = -Bv^2$.

- (i) Show that the distance (D_1) travelled in slowing the plane from speed V to speed U under the effect of air resistance only, is given by:

4

$$D_1 = \frac{M}{B} \ln\left(\frac{V}{U}\right)$$

After the brakes are applied, the plane experiences a constant resistive force of A Newtons (due to brakes) as well as a variable resistive force, Bv^2 . That is, $M\ddot{x} = -(A + Bv^2)$.

- (ii) After the brakes are applied when the plane is travelling at speed U , show that the distance D_2 required to come to rest is given by:

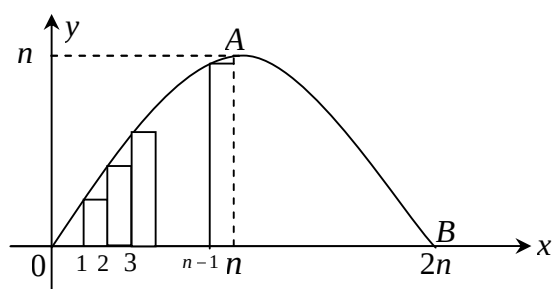
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$$D_2 = \frac{M}{2B} \ln \left[1 + \frac{B}{A} U^2 \right].$$

- (iii) Use the above information to estimate the total stopping distance after landing, for a 100 tonne plane if it slows from 90 m/s^2 to 60 m/s^2 under a resistive force of $125v^2$ Newtons and is finally brought to rest with the assistance of a constant braking force of magnitude 75 000 Newtons.

2

(b)



The diagram above represents the curve $y = n \sin \frac{\pi x}{2n}$, $0 \leq x \leq 2n$, where n is any integer $n \geq 2$.

The points $O(0,0)$, $A(n,n)$ and $B(2n,0)$ lie on this curve.

- (i) By considering the areas of the lower rectangles of width 1 from $x = 0$ to $x = n$, prove that

$$\sin \frac{\pi}{2n} + \sin \frac{2\pi}{2n} + \sin \frac{3\pi}{2n} + \dots + \sin \frac{\pi(n-1)}{2n} < \frac{2n}{\pi}.$$

3

- (ii) Hence or otherwise, explain why $2n \sum_{r=1}^{n-1} \sin \frac{\pi r}{2n} < \frac{\pi n^2}{2}$.

2

END OF PAPER

QUESTION 1

a) $\int \sin^3 \theta d\theta$

$$= \int \sin \theta (1 - \cos^2 \theta) d\theta$$

$$= \int (\sin \theta - \sin \theta \cos^2 \theta) d\theta$$

$$= -\cos \theta + \frac{\cos^3 \theta}{3} + C$$

b) (i) $a(x^2+1) + (bx+c)(x+1) = 3x+1$

Let $x = -1$, $2a + 0 = -2$

$$\therefore a = -1$$

coefficients of x^2 : $a + b = 0$

$$\therefore b = 1$$

Let $x = 0$ $\therefore -1 + c = 1$

$$\therefore c = 2$$

$$\therefore \frac{3x+1}{(x+1)(x^2+1)} = \frac{-1}{x+1} + \frac{x+2}{x^2+1}$$

(ii) $\int \frac{3x+1}{(x+1)(x^2+1)} dx = \int \left(\frac{-1}{x+1} + \frac{x+2}{x^2+1} \right) dx$

$$= -\ln(x+1) + \int \left(\frac{x}{x^2+1} + \frac{2}{x^2+1} \right) dx$$

$$= -\ln(x+1) + \frac{1}{2} \ln(x^2+1) + 2 \tan^{-1} x + C$$

c) $I = \int_1^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx$

$$x = 2 \sin \theta$$

$$\sqrt{4-x^2} = \sqrt{4-4\sin^2 \theta}$$

$$dx = 2 \cos \theta d\theta$$

$$= 2 \cos \theta$$

$$x = \sqrt{3} \quad \sin \theta = \frac{\sqrt{3}}{2}, \quad \theta = \frac{\pi}{3}$$

$$x = 1 \quad \sin \theta = \frac{1}{2}, \quad \theta = \frac{\pi}{6}$$

$$\therefore I = \int_{\pi/6}^{\pi/3} \frac{4 \sin^2 \theta \times 2 \cos \theta d\theta}{2 \cos \theta}$$

$$= \int_{\pi/6}^{\pi/3} 4 \sin^2 \theta d\theta$$

$$= 2 \int_{\pi/6}^{\pi/3} (1 - \cos 2\theta) d\theta$$

$$= \left[2\theta - \sin 2\theta \right]_{\pi/6}^{\pi/3}$$

$$= \frac{2\pi}{3} - \sin 2\pi/3 - \pi/3 + \sin \pi/3$$

$$= \pi/3$$

d) $I = \int x^2 \sqrt{3-x} dx$

$$u = 3-x$$

$$du = -dx$$

$$\therefore I = \int (3-u)^2 \sqrt{u} (-du)$$

$$= \int (9-6u+u^2) (-\sqrt{u}) du$$

$$= \int (-9\sqrt{u} + 6u^{3/2} - u^{5/2}) du$$

$$= -9u^{3/2} \times \frac{2}{3} + 6u^{5/2} \times \frac{2}{5} - u^{7/2} \times \frac{2}{7} + C$$

$$= -6(3-x)^{3/2} + \frac{12}{5}(3-x)^{5/2} - \frac{2}{7}(3-x)^{7/2} + C$$

e) $I = \int_0^1 \tan^{-1} x dx$

let $u = \tan^{-1} x$ $dV = dx$

$$du = \frac{1}{1+x^2} dx \quad V = x$$

$$\therefore I = \left[x \tan^{-1} x \right]_0^1 - \int_0^1 x \times \frac{1}{1+x^2} dx$$

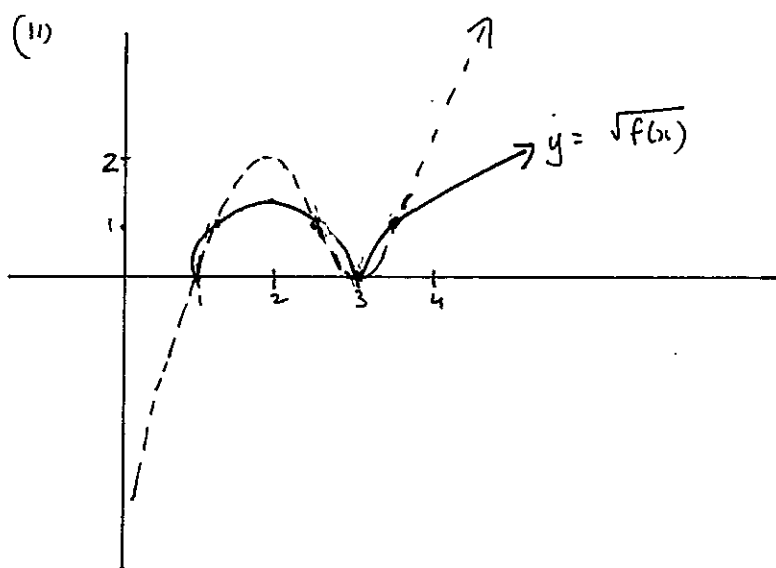
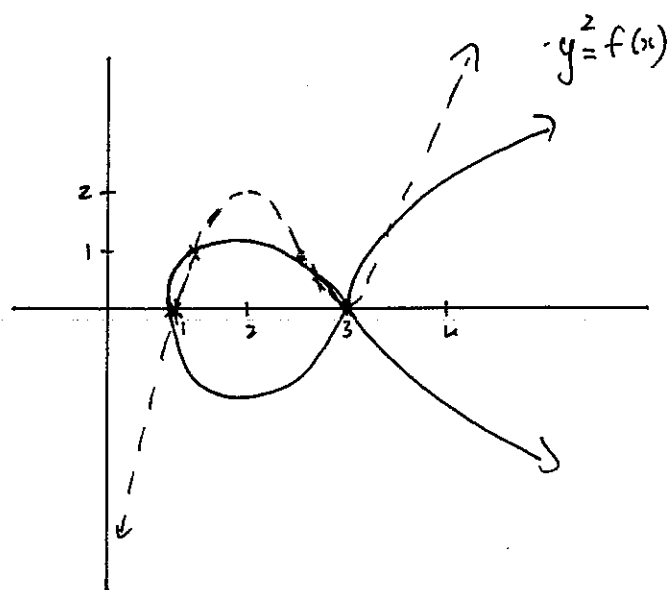
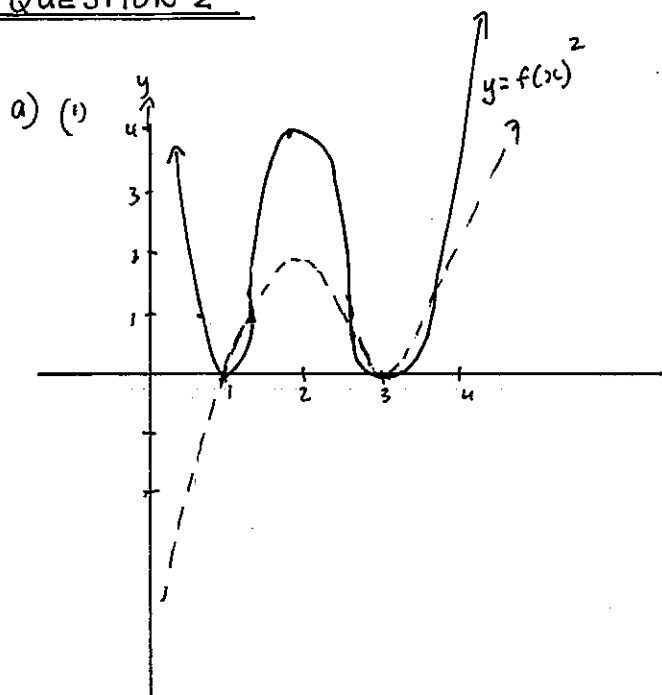
$$= \tan^{-1} 1 - 0 - \left[\frac{1}{2} \ln(1+x^2) \right]_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2 + \frac{1}{2} \ln 1$$

$$= \frac{\pi - \ln 4}{4}$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

QUESTION 2



b) (i) $f'(x) = \frac{2-x}{x^2} = \frac{2}{x^2} - \frac{1}{x}$

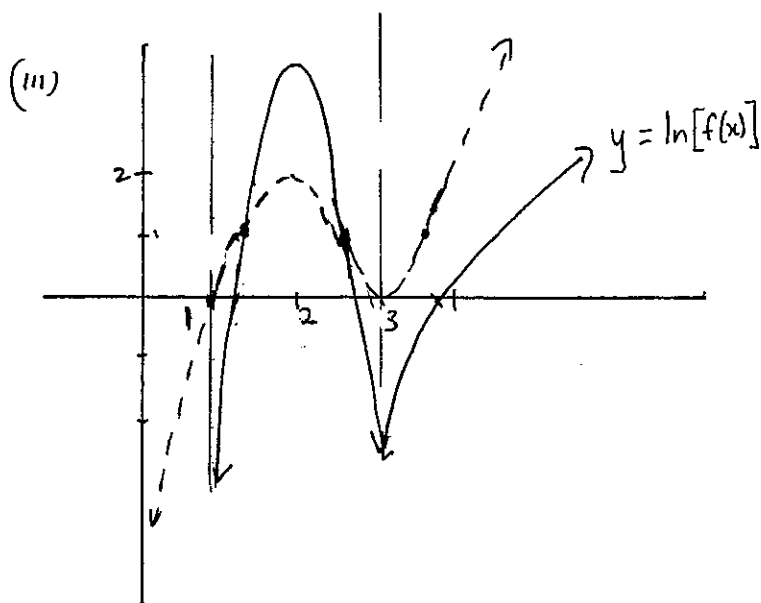
$$f(x) = \frac{2x^{-1}}{-1} - \ln x + C$$

$$f(1) = 0$$

$$0 = -2 - \ln 1 + C \Rightarrow C = 2$$

$$\therefore f(x) = -\frac{2}{x} - \ln x + 2$$

$$f''(x) = -4x^{-3} + x^{-2} = -\frac{4}{x^3} + \frac{1}{x^2}$$



(ii) Stationary points at $f'(x) = 0$

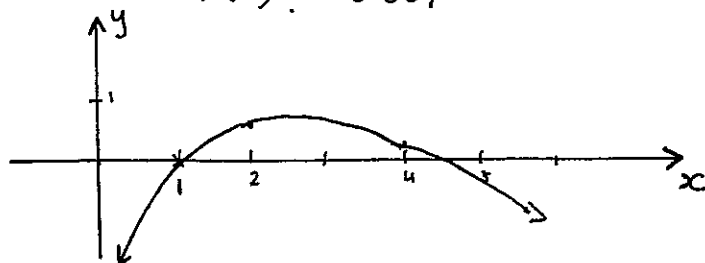
$$\therefore \text{at } 2-x=0 \Rightarrow x=2$$

$$\text{at } x=2, y = -1 - \ln 2 + 2 = 1 - \ln 2$$

$$f''(x) = -\frac{4}{x^3} + \frac{1}{x^2} = -\frac{1}{4} < 0 \therefore \text{max tp at } (2, 1 - \ln 2)$$

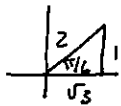
(iii) $f(4) \doteq 0.1137$

$$f(5) \doteq -0.009$$



QUESTION 3

$$a) (\sqrt{3} + i)^8 = (2 \operatorname{cis} \pi/6)^8$$



$$= 2^8 \operatorname{cis} \frac{8\pi}{6}$$

$$= 256 \operatorname{cis} \left(-\frac{2\pi}{3}\right)$$

$$= 256 \left(\cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right) \right)$$

$$= 256 \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right)$$

$$= 256 \left(-\frac{1}{2} - i \times \frac{\sqrt{3}}{2} \right)$$

$$= -128 - 128\sqrt{3}i$$

$$d) (i) \frac{-1+i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i} = \frac{-\sqrt{3}+i+i\sqrt{3}+1}{3+1}$$

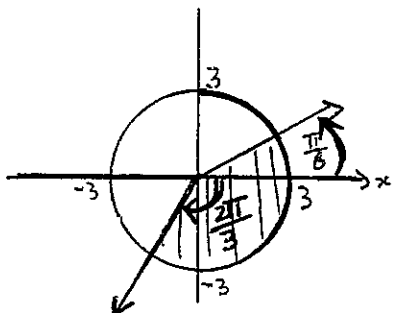
$$= \frac{1-\sqrt{3}+i(1+\sqrt{3})}{4}$$

$$\therefore \frac{1-\sqrt{3}}{4} + \frac{i(1+\sqrt{3})}{4} = \frac{1}{\sqrt{2}} \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right)$$

$$\therefore \frac{1-\sqrt{3}}{4} = \frac{1}{\sqrt{2}} \cos \frac{7\pi}{12}$$

$$\cos \frac{7\pi}{12} = \frac{\sqrt{2}-\sqrt{6}}{4}$$

b)



$$c) LHS = \frac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta} \times \frac{1+\sin\theta+i\cos\theta}{1+\sin\theta+i\cos\theta}$$

$$= \frac{(1+\sin\theta)^2 + 2i\cos\theta(1+\sin\theta) + (i\cos\theta)^2}{(1+\sin\theta)^2 + (\cos\theta)^2}$$

$$= \frac{1+2\sin\theta+\sin^2\theta+2i\cos\theta+2i\cos\theta\sin\theta-\cos^2\theta}{1+2\sin^2\theta+\sin^2\theta+\cos^2\theta}$$

$$= \frac{2\sin^2\theta+2\sin\theta+2i\cos\theta\sin\theta+2i\cos\theta}{2+2\sin\theta}$$

$$= \frac{2\sin\theta(\sin\theta+1)+2i\cos\theta(\sin\theta+1)}{2(1+\sin\theta)}$$

$$= \frac{2(\sin\theta+i\cos\theta)(\sin\theta+1)}{2(1+\sin\theta)}$$

$$= \sin\theta+i\cos\theta$$

$$= RHS$$

$$e) z_2 = i z_1$$

$$LHS = (z_1 + z_2)^2$$

$$= z_1^2 + 2z_1 z_2 + z_2^2$$

$$= z_1^2 + 2z_1 \times i z_1 + (i z_1)^2$$

$$= z_1^2 + 2i z_1^2 - z_1^2$$

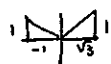
$$= 2i z_1^2$$

$$= 2z_1 \times i z_1$$

$$= 2z_1 z_2$$

$$= RHS$$

$$d) (i) z = \frac{-1+i}{\sqrt{3}+i}$$



$$= \frac{\sqrt{2} \operatorname{cis} \left(\frac{3\pi}{4}\right)}{2 \operatorname{cis} \pi/6}$$

$$= \frac{\sqrt{2}}{2} \times \frac{\sqrt{2}}{2} \operatorname{cis} \left(\frac{7\pi}{12}\right)$$

$$= \frac{1}{2} \operatorname{cis} \frac{7\pi}{12}$$

QUESTION 4

a) If $z = 1+i$ is a root of $P(z)$

then $z = 1-i$ is also a root.

$$\begin{aligned} \therefore z^3 + pz^2 + qz + 6 &= (z-1-i)(z-1+i)(z-a) \\ &= (z^2 - 2z + 1 + 1)(z-a) \\ &= (z^2 - 2z + 2)(z-a) \end{aligned}$$

equating constant terms, $a = -3$

$$\begin{aligned} \therefore P(z) &= (z^2 - 2z + 2)(z + 3) \\ &= z^3 + z^2 - 4z + 6 \end{aligned}$$

$$\therefore p = 1 \text{ and } q = -4$$

b) $f(x) = x^3 + px + q$

$$f'(x) = 3x^2 + p$$

If $f(x)$ has a multiple root then $f'(x) = 0$

$$3x^2 + p = 0$$

$$x^2 = -\frac{p}{3}$$

$$\therefore f(x) = x(x^2 + p) + q = 0$$

$$x(-\frac{p}{3} + p) + q = 0$$

$$x \times \frac{2p}{3} + q = 0$$

$$x = -\frac{3q}{2p}$$

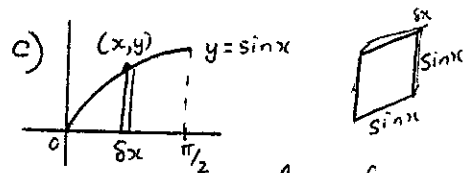
$$\therefore f(x) = \left(-\frac{3q}{2p}\right)^3 + p\left(-\frac{3q}{2p}\right) + q = 0$$

$$\frac{-27q^3}{8p^3} - \frac{3q}{2} + q = 0$$

$$-27q^3 - 12p^3q + 8p^3q = 0$$

$$27q^3 + 4p^3q = 0$$

$$4p^3 + 27q^2 = 0$$



Area of cross-section = $\sin^2 x$

$\therefore$ Volume of cross-section = $\sin^2 x \delta x$

$$\text{Vol. of solid} = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\pi/2} \sin^2 x \delta x$$

$$= \int_0^{\pi/2} \sin^2 x \, dx$$

$$= \frac{1}{2} \int_0^{\pi/2} (1 - \cos 2x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left(\frac{\pi}{2} - \frac{\sin \pi}{2} \right) - \frac{1}{2} \left(0 - \frac{\sin 0}{2} \right)$$

$$= \frac{\pi}{4} \text{ cubic units}$$

d) (i) Let $z = \cos \theta + i \sin \theta$

$$\therefore z^5 = \cos 5\theta + i \sin 5\theta = 1$$

$$5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$$

$$\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$$

$\therefore$ Roots are $z_1 = \cos 0 + i \sin 0 (=1)$

$$z_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$z_3 = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$$

$$z_4 = \cos(-\frac{2\pi}{5}) + i \sin(-\frac{2\pi}{5})$$

$$z_5 = \cos(-\frac{4\pi}{5}) + i \sin(-\frac{4\pi}{5})$$

$$(ii) z^5 - 1 = (z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)$$

$$= (z-1)(z^2 - z(z_2 + \bar{z}_2) + z_2\bar{z}_2)(z^2 - z(z_3 + \bar{z}_3) + z_3\bar{z}_3)$$

$$z_2 + \bar{z}_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}$$

$$= 2 \cos \frac{2\pi}{5}$$

$$z_2\bar{z}_2 = (\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5})(\cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5})$$

$$= \cos^2 \frac{2\pi}{5} + \sin^2 \frac{2\pi}{5}$$

$$= 1$$

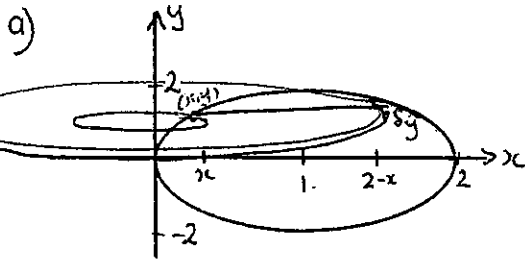
$$\therefore z^5 - 1 = (z-1)(z^2 - 2 \cos \frac{2\pi}{5} z + 1)(z^2 - 2 \cos \frac{4\pi}{5} z + 1)$$

(iii) Sum of roots = $-\frac{b}{a}$

$$\therefore 1 + 2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5} = 0$$

$$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

QUESTION 5



$$\begin{aligned}\text{Area of slice} &= \pi R^2 - \pi r^2 \\ &= \pi(R-r)(R+r)\end{aligned}$$

$$(x-1)^2 + \frac{y^2}{4} = 1$$

$$(x-1)^2 = \frac{4-y^2}{4}$$

$$x = 1 \pm \frac{\sqrt{4-y^2}}{2}$$

$$R-r = \sqrt{4-y^2} \quad R+r = 2$$

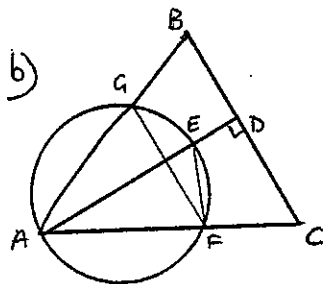
$$\begin{aligned}\text{Area of slice} &= \pi(\sqrt{4-y^2}) \times 2 \\ &= 2\pi\sqrt{4-y^2}\end{aligned}$$

$$\text{Vol of slice} = 2\pi\sqrt{4-y^2} \delta y$$

$$\begin{aligned}\text{Vol of solid} &= \lim_{\delta y \rightarrow 0} \sum_{y=-2}^2 2\pi\sqrt{4-y^2} \delta y \\ &= 2\pi \times 2 \int_0^2 \sqrt{4-y^2} dy\end{aligned}$$

Now $\int_0^2 \sqrt{4-y^2} dy = \frac{\pi \times 2^2}{4}$ since it represents the area of a quarter circle, radius 2 units.

$$\begin{aligned}\therefore \text{Vol of solid} &= 2\pi \times 2 \times \frac{\pi}{2} \\ &= 4\pi^2 \text{ cubic units.}\end{aligned}$$



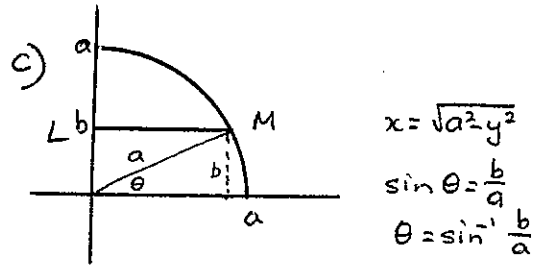
$$\angle AGF = \angle AEF \quad (\angle^s \text{ in same segment})$$

EDFC is a cyclic quad with diameter EC since $\angle EDC = 90^\circ$

$$\therefore \angle AEF = \angle DCF \quad (\text{ext. } \angle \text{ of a cyclic quad.})$$

$$\therefore \angle AGF = \angle DCF$$

$\therefore$ GBFC is a cyclic quad since its ext. $\angle$ is equal to the opposite interior $\angle$.



$$\begin{aligned}x &= \sqrt{a^2 - y^2} \\ \sin \theta &= \frac{b}{a} \\ \theta &= \sin^{-1} \frac{b}{a}\end{aligned}$$

$$(i) \text{ Area of quadrant} = \frac{\pi a^2}{4}$$

$$\therefore \text{Area of lower half} = \frac{\pi a^2}{8}$$

= sector + triangle

$$\therefore \frac{1}{2} a^2 \theta + \frac{1}{2} b \sqrt{a^2 - b^2} = \frac{\pi a^2}{8}$$

$$a^2 \sin^{-1} \frac{b}{a} + b \sqrt{a^2 - b^2} = \frac{\pi a^2}{4}$$

$$\sin^{-1} \frac{b}{a} + \frac{b}{a^2} \sqrt{a^2 - b^2} = \frac{\pi}{4}$$

(ii) If $a=1$, then

$$\sin^{-1} b + b \sqrt{1-b^2} = \frac{\pi}{4}$$

$$\theta + \sin \theta \sqrt{1-\sin^2 \theta} = \frac{\pi}{4}$$

$$\theta + \sin \theta \times \cos \theta = \frac{\pi}{4}$$

$$\theta + \frac{1}{2} \sin 2\theta = \frac{\pi}{4}$$

$$\therefore \sin 2\theta = \frac{\pi}{2} - 2\theta$$

(iii) Could use Newton's method to solve this equation.

(or halving the interval method

$$\text{OR graph } y = \sin 2\theta + y = \frac{\pi}{2} - 2\theta$$

and find their points of intersection).

QUESTION 6

$$a) \frac{x^2}{4} + \frac{y^2}{3} = 1$$

$$a=2, b=\sqrt{3}$$

$$(i) b^2 = a^2(1-e^2)$$

$$3 = 4(1-e^2)$$

$$e = \frac{1}{2}$$

$$\text{Foci at } (\pm ae, 0) = (\pm 1, 0)$$

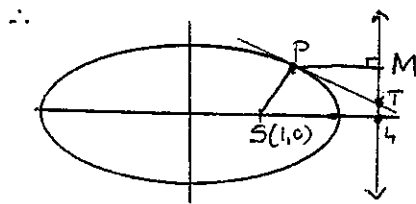
$$\text{Directrices } x = \pm \frac{a}{e}$$

$$= \pm \frac{2}{\frac{1}{2}}$$

$$\therefore x = \pm 4$$

$$(ii) P(x_1, y_1)$$

By definition, $SP = e \times PM$



$$\therefore SP = e \times PM$$

$$S'P = e \times PM'$$

$$SP + S'P = e(PM + PM')$$

$$= \frac{1}{2}(MM')$$

$$= \frac{1}{2} \times 8$$

$$= 4 \text{ which is a constant}$$

$\therefore SP + S'P$ is independent of the position of P.

$$(iii) \frac{2x}{4} + \frac{2y}{3} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-3x}{4y}$$

$$\text{At } P, \text{ m of tangent} = \frac{-3x_1}{4y_1}$$

$$\therefore \text{Eq'n of tangent is } y - y_1 = \frac{-3x_1}{4y_1}(x - x_1)$$

$$4yy_1 - 4y_1^2 = -3xx_1 + 3x_1^2$$

$$3xx_1 + 4yy_1 = 3x_1^2 + 4y_1^2$$

$$\frac{xx_1}{4} + \frac{yy_1}{3} = \frac{x_1^2}{4} + \frac{y_1^2}{3}$$

$$\therefore \frac{xx_1}{4} + \frac{yy_1}{3} = 1$$

$$(iv) \text{ At } T, x=4 \therefore x_1 + \frac{yy_1}{3} = 1$$

$$\therefore T(4, \frac{3}{y_1}(1-x_1)), S(1,0)$$

If $\angle PST$ is a right $\angle$, $m_{PS} \times m_{ST} = 1$

$$\text{LHS} = \frac{y_1}{x_1-1} \times \frac{\frac{3}{y_1}(1-x_1)}{4-1}$$

$$= \frac{y_1}{x_1-1} \times \frac{-3}{y_1} \frac{(x_1-1)}{3}$$

$$= -1$$

$$= \text{RHS}$$

$\therefore \angle PST$ is a right angle

$$b) (i) (a-b)^2 \geq 0$$

$$a^2 - 2ab + b^2 \geq 0$$

$$\therefore a^2 + b^2 \geq 2ab$$

$$(ii) \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

$$= \frac{ab+ac+bc}{abc}$$

$$= \frac{(ab+ac+bc)(a+b+c)}{abc} \quad \text{since } a+b+c=1$$

$$= \frac{a^2b + a^2c + abc + ab^2 + abc + b^2c + abc + ac^2 + bc^2}{abc}$$

$$= \frac{3abc + c(a^2+b^2) + a(b^2+c^2) + b(a^2+c^2)}{abc}$$

$$> \frac{3abc + c \times 2ab + a \times 2bc + b \times 2ac}{abc} \quad \text{from (i)}$$

$$= \frac{9abc}{abc}$$

$$= 9 \quad \therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$$

$$(iii) (1-a)(1-b)(1-c)$$

$$= (b+c)(a+c)(a+b) \quad \text{since } a+b+c=1$$

$$= (b+c)(a^2+ab+ac+bc)$$

$$= a^2b + ab^2 + abc + b^2c + a^2c + abc + ac^2 + bc^2$$

$$= b(a^2+c^2) + a(b^2+c^2) + c(b^2+a^2) + 2abc$$

$$> b \times 2ac + a \times 2bc + c \times 2ab + 2abc$$

$$= 8abc$$

$$\therefore (1-a)(1-b)(1-c) \geq 8abc$$

QUESTION 7

a) $xy = c^2 \quad T(ct, \frac{c}{t})$

tangent at T: $x + t^2 y = 2ct$

(i) $P(2ct, 0)$

$Q(0, \frac{2c}{t})$

(ii) at T, $m_{\tan} = -\frac{1}{t^2} \therefore m_{\text{Norm}} = t^2$

$\therefore$ eq'n of normal:

$y - \frac{c}{t} = t^2(x - ct)$

$y = t^2 x - ct^3 + \frac{c}{t}$

(iii) at R, $y = x$

$\therefore x = t^2 x - ct^3 + \frac{c}{t}$

$x(t^2 - 1) = c(t^3 - \frac{1}{t})$

$x(t^2 - 1) = \frac{c}{t}(t^4 - 1)$

$x = \frac{c}{t} \frac{(t^4 - 1)}{(t^2 - 1)}$

$\therefore x = \frac{c}{t}(t^2 + 1)$

(iv) $\therefore R(\frac{c}{t}(t^2 + 1), \frac{c}{t}(t^2 + 1))$

If $\triangle OTR$ is isosceles,

$OT = TR$

LHS = $\sqrt{c^2 t^2 + \frac{c^2}{t^2}}$

$= \sqrt{c^2(t^2 + \frac{1}{t^2})}$

RHS = $\sqrt{(ct - \frac{c}{t}(t^2 + 1))^2 + (\frac{c}{t} - \frac{c}{t}(t^2 + 1))^2}$

$= \sqrt{c^2(t - t + \frac{1}{t})^2 + c^2(\frac{1}{t} - t - \frac{1}{t})^2}$

$= \sqrt{c^2(\frac{1}{t^2} + t^2)}$

$= \text{LHS}$

$\therefore \triangle OTR$ is isosceles

b) $I_n = \int \frac{dx}{(x^2 + 1)^n} \quad \text{let } U = (x^2 + 1)^{-n}$

(i) $du = -n \times 2x(x^2 + 1)^{-n-1} dx$

$dV = dx$

$\therefore V = x$

$\therefore I_n = x(x^2 + 1)^{-n} - \int x \times -n \times 2x(x^2 + 1)^{-n-1} dx$

$= \frac{x}{(x^2 + 1)^n} + 2n \int \frac{x^2 dx}{(x^2 + 1)^{n+1}}$

$= \frac{x}{(x^2 + 1)^n} + 2n \int \frac{x^2 + 1 - 1}{(x^2 + 1)^{n+1}} dx$

$= \frac{x}{(x^2 + 1)^n} + 2n \int \frac{dx}{(x^2 + 1)^n} - 2n \int \frac{dx}{(x^2 + 1)^{n+1}}$

$\therefore I_n = \frac{x}{(x^2 + 1)^n} + 2n I_n - 2n I_{n+1}$

$2n I_{n+1} = I_n (2n - 1) + \frac{x}{(x^2 + 1)^n}$

putting $n+1 = n, \Rightarrow n = n-1$

$2(n-1) I_n = I_{n-1} (2(n-1) - 1) + \frac{x}{(x^2 + 1)^{n-1}}$

$I_n = \frac{1}{2(n-1)} \left[\frac{x}{(x^2 + 1)^{n-1}} + (2n-3) I_{n-1} \right]$

(ii) $\int_0^1 \frac{dx}{(x^2 + 1)^2} = I_2$

$\therefore I_2 = \frac{1}{2} \left[\left(\frac{x}{x^2 + 1} \right)' + 1 \times I_1 \right]$

$= \frac{1}{2} \left[\frac{1}{2} - 0 + \int_0^1 \frac{dx}{x^2 + 1} \right]$

$= \frac{1}{4} + \left[\frac{1}{2} \tan^{-1} x \right]_0^1$

$= \frac{1}{4} + \frac{1}{2} \times [\tan^{-1} 1 - \tan^{-1} 0]$

$= \frac{1}{4} + \frac{1}{2} \times \frac{\pi}{4}$

$= \frac{\pi + 2}{8}$

QUESTION 8

a) (i) $M \ddot{x} = -Bv^2$

$$v \frac{dv}{dx} = \frac{-Bv^2}{M}$$

$$\frac{dv}{dx} = \frac{-Bv}{m}$$

$$\frac{dx}{dv} = \frac{-m}{Bv}$$

$$x = \int_U^v \frac{-m}{Bv} dv$$

$$\therefore D_1 = \left[\frac{-m}{B} \ln v \right]_V^U$$

$$= -\frac{m}{B} \ln U + \frac{m}{B} \ln V$$

$$D_1 = \frac{m}{B} \ln \left(\frac{V}{U} \right)$$

(ii) $M \ddot{x} = -(A+Bv^2)$

$$v \frac{dv}{dx} = \frac{-(A+Bv^2)}{M}$$

$$\frac{dv}{dx} = \frac{-(A+Bv^2)}{Mv}$$

$$\frac{dx}{dv} = \frac{-Mv}{A+Bv^2}$$

$$x = M \int_U^0 \frac{-v}{A+Bv^2} dv$$

$$\therefore D_2 = \left[-\frac{M}{2B} \ln(A+Bv^2) \right]_U^0$$

$$= -\frac{M}{2B} \ln A + \frac{M}{2B} \ln(A+BU^2)$$

$$= \frac{M}{2B} \ln \left(\frac{A+BU^2}{A} \right)$$

$$D_2 = \frac{M}{2B} \ln \left(1 + \frac{B}{A} U^2 \right)$$

(iii) $M = 100000 \text{ kg}, V = 90, U = 60$

$$Bv^2 = 125v^2 \therefore B = 125, A = 75000$$

$$\therefore \text{total distance} = D_1 + D_2$$

$$= \frac{100000}{125} \ln \left(\frac{90}{60} \right) + \frac{100000}{250} \ln \left(1 + \frac{125}{75000} \times 60^2 \right)$$

$$= 800 \ln 1.5 + 400 \ln (7)$$

$$= 1102.736 \dots$$

$$\approx 1103 \text{ metres.}$$

b) (i) By adding area of rectangles,

A_1 = area under curve

$$\approx n \sin \frac{\pi}{2n} + n \sin \frac{2\pi}{2n} + n \sin \frac{3\pi}{2n} + \dots + n \sin \frac{(n-1)\pi}{2n}$$

By integrating to find the area under curve,

$$A_2 = \int_0^n n \sin \frac{\pi x}{2n} dx$$

$$= \left[n \times \frac{2n}{\pi} \times -\cos \frac{\pi x}{2n} \right]_0^n$$

$$= -\frac{2n^2}{\pi} \cos \frac{n\pi}{2n} + \frac{2n^2}{\pi} \cos 0$$

$$= -\frac{2n^2}{\pi} \cos \frac{\pi}{2} + \frac{2n^2}{\pi} \cos 0$$

$$= \frac{2n^2}{\pi} (0+1)$$

$$= \frac{2n^2}{\pi}$$

$$A_1 < A_2$$

$$\therefore n \left(\sin \frac{\pi}{2n} + \sin \frac{2\pi}{2n} + \sin \frac{3\pi}{2n} + \dots + \sin \frac{\pi(n-1)}{2n} \right) < \frac{2n^2}{\pi}$$

$$\therefore \sin \frac{\pi}{2n} + \sin \frac{2\pi}{2n} + \sin \frac{3\pi}{2n} + \dots + \sin \frac{\pi(n-1)}{2n} < \frac{2n}{\pi}$$

(ii) From (i) $\sum_{r=1}^{n-1} \sin \frac{r\pi}{2n} < \frac{2n}{\pi}$

$$\therefore 2n \sum_{r=1}^{n-1} \sin \frac{r\pi}{2n} < \frac{4n^2}{\pi} = \frac{4\pi n^2}{\pi^2}$$

$$\text{but } \frac{4}{\pi} < \frac{\pi}{2} \quad (\text{since } 8 < \pi^2)$$

$$\therefore \frac{4\pi n^2}{\pi \times \pi} < \frac{\pi}{2} \times \frac{\pi n^2}{\pi}$$

$$= \frac{\pi n^2}{2}$$

$$\therefore 2n \sum_{r=1}^{n-1} \sin \frac{r\pi}{2n} < \frac{\pi n^2}{2}$$